

Assessing Boltz-2 Performance for the Binding Classification of Docking Hits

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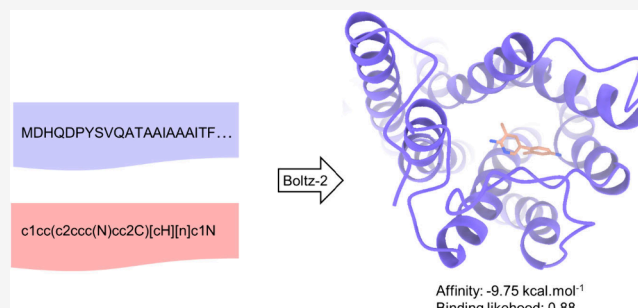


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ABSTRACT: The recently released Boltz-2 cofolding model is generating high expectations by enabling both protein–ligand structure and binding affinity predictions. When applied to a recently described and challenging data set of ultralarge-virtual-screening hits, Boltz-2 excels at discriminating true from false positives, overcoming by a large margin all scoring functions tested so far on raw docking poses. Strikingly, affinity predictions seem to be relatively independent of pose quality but are not biased by obvious chemical similarity to known compounds sharing comparable binding potencies. To ascertain that Boltz-2 truly relies on the physics of intermolecular interactions, we challenged affinity predictions with biologically meaningful challenges (target mutation and target shuffling). Binary classification of active vs inactive compounds remains insensitive to key binding site mutations and even in some cases to target exchange, raising concerns on the hidden features governing Boltz-2 affinity predictions.



INTRODUCTION

Among the many promises of artificial intelligence (AI) to facilitate early drug discovery, protein–ligand cofolding methods arouse enormous interest from the drug design community.¹ The beauty of cofolding lies in the possibility of predicting a protein–ligand three-dimensional (3D) structure from the simple knowledge of a target's amino acid sequence and a simple 1D ligand string. Relying on the spectacular accuracy of deep learning methods such as AlphaFold2 (AF2)² to predict protein structures, cofolding exemplified by its AlphaFold3 (AF3)³ upgrade introduced a diffusion-based generative model to predict biomolecular interactions with other macromolecules (proteins, nucleic acids) and ligands (cofactors, drug candidates). Since usage of the AF3 model, notably the release of weights, has been strictly controlled, open-source and web-based alternatives have been flourishing in recent months, including Boltz-1,⁴ Chai-1,⁵ NeuralPLexer3,⁶ Protenix,⁷ RosettaFold All-Atom,⁸ and UMol.⁹ Such models enable, in favorable cases (well-known folds, targets, and binding sites), prediction of protein–ligand structures at good resolution. Predicting the binding strength of a ligand to its target still relies on third-party software, notably free energy perturbation (FEP), for the most accurate but time-consuming approach.¹⁰ A possible game-changer was recently described by the MIT Jamel Clinic and Recursion Pharmaceuticals, with the release of Boltz-2, the first AI framework able to simultaneously predict both the structure and binding affinity of protein–ligand complexes.¹¹ Boltz-2 notably uses a Pairformer architecture, which is an iterative refinement module that processes and updates pairwise representations, essentially a

2D matrix where each entry represents the relationships or distance information between a pair of tokens (e.g., amino acid residues, atoms, or small molecules) in the input. Boltz-2 produces 3 outputs: (i) a 3D structure of the protein–ligand complex in mmCIF or PDB format, (ii) a binding probability value between 0 and 1 (classification), and (iii) a binding free energy (binding affinity) prediction (regression) in pIC₅₀ unit. In this study, we refer to these outputs as B2-S, B2-C, and B2-A, for structure, classification, and binding affinity prediction, respectively (additional details about the training of Boltz-2 are provided in [Computational Methods](#)). By training on large sets of protein structures (X-ray diffraction, cryo-electron microscopy, NMR ensembles, molecular dynamics snapshots) as well as curated binding affinity data (binary and continuous labels) from public sources,^{12–14} Boltz-2 predicts structures with the accuracy of AF3 while predicting affinities at a claimed FEP accuracy but 1,000 times faster.¹¹

Over recent months, cofolding models have attracted much attention in peer-reviewed articles,^{15–17} preprints, and blogs,^{18,19} attempting to better delineate their true applicability domain. While very accurate at predicting structures and affinities from complexes between rigid proteins and known

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chemotypes, they still suffer from the lack of reliable training data to predict truly novel protein–ligand structures,¹⁸ to account for significant ligand-induced conformational changes,¹⁵ or to pose allosteric modulators in the right pocket.¹⁶ Moreover, cofolding models cannot yet take into account the crucial role of cofactors, ions, and water molecules.¹⁹ Intriguingly, predicting binding affinities for a small series of congeneric ligands seems somehow disconnected from the posing accuracy of Boltz-2.¹⁸ This observation suggests that cofolding methods may be memorizing more than really learning the physics of molecular interactions. Whereas most benchmarks have been focusing on the accuracy of cofolded structures, Boltz-2 binding affinity predictions still need to be properly challenged. Such predictions may be evaluated in two different contexts using two separate models. On the one hand, binary discrimination of binders vs nonbinders, addressed by a simple classifier outputting the binding probability, is clearly relevant in hit identification. On the other hand, a regressor trained on affinity data may be applied in a hit-to-lead optimization scenario.¹¹

We herewith challenge both the classifier and the regressor models with a previously reported set of 943 docking-based ultralarge-virtual-screening hits (ULVSH) covering ten targets, all being annotated with experimental *in vitro* data (binding and/or functional assays).²⁰ The data set is quite challenging for two main reasons: (i) each hit list has been very carefully designed by independent groups and docking/postdocking workflows and contains the top-scoring results among several hundred million screened compounds;^{21–30} (ii) none of the many tested affinity predictors (empirical scoring functions, machine/deep learning, molecular mechanics, semiempirical quantum mechanics) has been able yet to properly discriminate true from false positives from this data set.²⁰ Because of the very intriguing outcome of Boltz-2 predictions on this particular data set, potential biases have been searched and adversarial challenges have been designed to better assess the true value of affinity predictions generated by this novel structural biology foundation model.

■ COMPUTATIONAL METHODS

Ultralarge-Virtual-Screening Hits Data Set (ULVSH)

The ULVSH data set²⁰ consists of ten targets for which 943 chemically diverse compounds, selected by binding site-tailored docking and carefully postprocessed, were tested experimentally for *in vitro* binding and/or functional assays, yielding 364 true positives and 579 false positives. All 943 compounds are primary virtual screening hits from ultralarge docking screens (74 million to 1.2 billion on-demand compounds from Enamine REAL space) that were remarkably successful with respect to hit rate (>20%). These hits originate from independent studies and research groups, were tested at a single concentration (usually 10 μ M) in duplicates, and are specific to each of the ten targets.

All true positives (binders) have been confirmed by dose–response curves in duplicate or triplicate to determine inhibition constants (for eight targets) or EC₅₀ values (CASR and MTR1A sets). For most targets, affinities of true positives are spread over 2–4 log pK units and centered on a 1 μ M value (Figure S1). In a drug discovery pipeline, binders would be labeled as hits. With respect to target and fold space, there is a clear low diversity (70% of targets are GPCRs), although binding site diversity is clearly larger (only CNR1/CNR2 and DRD3/DRD4 binding site pairs exhibit significant sequence and structural similarity).

Raw docking poses (MOL2 or SD file formats) of all compounds were converted to SMILES strings using Openbabel v3.1.1.³¹ Targets were described by their amino acid sequence (ADRA2B: P18089,

CASR: P41180, CNR1: P21554, CNR2: P34972, DRD3: P35462, DRD4: P21917, MTR1A: P48039, ROCK1: Q13464, SC6A4: P31645, SGMR2: Q5BJF2) fetched from the UniProt knowledge-base.³² Please note that the sequence of human SGMR2 was used here, although the original ultralarge virtual screen was performed on the structure from the bovine receptor,²¹ to which it shares 100% binding site sequence identity.

Boltz-2 Cofolding

Boltz-2 (v.2.2.0) was downloaded from the Github repository³³ and used with default settings using as sole inputs YAML configuration files describing the target sequence and the compound SMILES string. Boltz-2 has been trained on both structures (proteins, nucleic acids, and complexes with pharmacological ligands) and affinity data (continuous and binary) to produce three separate models that were used without any further training. The first one (B2-S) predicts the 3D structure of the protein–ligand complexes (by default, one pose per compound) in the PDB format. The second (B2-A) and third (B2-C) models predict binding affinity and binding probability values, respectively. Boltz-2 has been trained on ca. 205,000 protein structures (X-ray, cryo-EM, NMR) from the RSCB-PDB, including 45,000 pharmacological ligands for which ten conformers are generated. In addition, approximately 15,000 molecular dynamics (MD) snapshots of diverse proteins are used to account for target flexibility. Last, an unknown number of predicted structures from the previous Boltz-1 model (RNAs, protein–DNA, RNA–ligand, protein–ligand, TCR–pMHC complexes) and from the AlphaFold database (5 million unique monomeric structures) were added as “distillation sets” to expand the volume of training data.

To predict binding affinity and binding probability, Boltz-2 has been trained on 1.2 million experimental affinity values from various sources (ChEMBL,¹² BindingDB¹⁴) and 200,000 binary data (binder vs nonbinder) from PubChem.³⁴ The binary classification data set is augmented by generating synthetic decoys (one per active) created by shuffling binders identified in hit-to-lead screens across different targets. For more details, the reader is referred to the original Boltz-2 preprint.¹¹ All predictions were computed with default Boltz-2 models on an NVIDIA Tesla V100 SXM2 32 GB GPU.

Pairwise Similarity Calculations

2.4 million experimentally tested compounds were downloaded as SMILES strings from ChEMBL v.34¹² and ionized at pH 7.4 as previously described.³⁵ Two-dimensional (2D) similarity between ULVSH and ChEMBL compounds was computed from Morgan-2 fingerprints (radius = 2, length = 2,048 bits) in RDKit v.2021.09.4³⁶ and estimated by the Tanimoto coefficient. Since Boltz-2 has been trained on both true actives and synthetic decoys, all ChEMBL compounds, regardless of their label and available binding data, were kept for comparison.

In addition, the same pairwise similarity analysis was performed with OpenEye's ROCS v.3.6.2³⁷ to account for 3D similarity (shape, pharmacophoric color), restricting here the comparison to ChEMBL compounds sharing the same targets as ULVSH hits. Whereas docking poses were directly used as queries, ChEMBL compounds were first converted into 3D coordinates from 2D SD to MOL2 file format using Corina v.3.40.³⁸ A maximum of 250 conformers per compound was then generated using standard settings of OpenEye's Omega2 v5.1.³⁷ Lastly, each query was aligned with each ChEMBL conformer with standard settings of ROCS. 3D similarity was expressed by the TanimotoCombo coefficient, the higher value being kept for each query.

Data Analysis

Area under the ROC curves (ROC AUC) were computed with PipelinePilot v.22.1.0.³⁹ Root-mean square deviations (RMSDs) were computed on heavy atoms using the obrms routine of Open Babel.³¹ Ligand RMSD calculations were considered between Boltz-2 and the original raw docking pose. Binding site RMSD calculations were computed between the Boltz-2 predicted structure and the original 3D structure used for virtual screening, considering only amino acids for which at least one heavy atom was less than 5.0 Å away from any

heavy atom of all docked compounds for the corresponding target. The local distance difference test (LDDT) between Boltz-2 and original docking protein–ligand structures was applied to the predicted protein–ligand interface by computing the LDDT-PLI score with OpenStructure v.2.11.0,⁴⁰ as recently described.⁴¹

RESULTS AND DISCUSSION

Boltz-2 Succeeds in Discriminating True from False Positives

All 943 protein–ligand complexes from the ULVSH set were cofolded with Boltz-2 with an average wall-clock time of 2 min

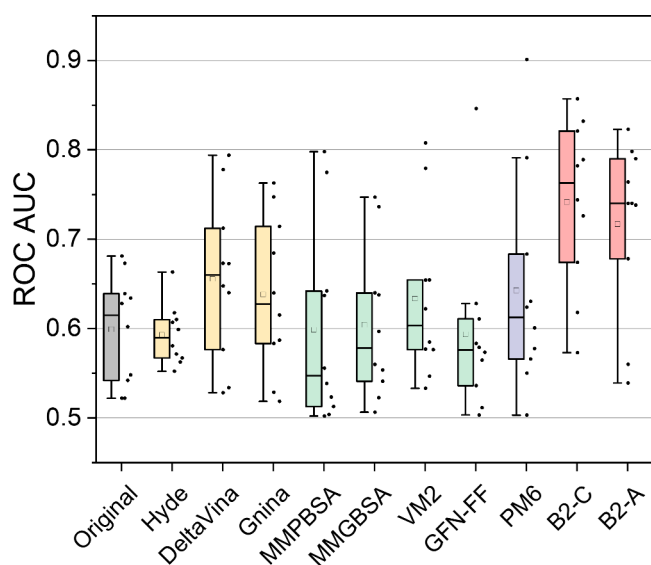


Figure 1. Area under the ROC curves (ROC AUC) for classifying true active and true inactive compounds for ten target sets from docking poses. Raw docking scores (DOCK, ICM-Pro, FRED, GOLD-PLP) used to prioritize ULVSH compounds have been assigned to the “Original” method. Predictions for all scoring functions were taken from a previous study.²⁰ Predictions from Boltz-2 are labeled “B2-C” for binding probability and “B2-A” for binding affinity. The boxes delimit the first and the third quartiles, and the whiskers delimit the 1.5 standard deviation. The median and the mean scores are indicated by a horizontal line and a square located in each box, respectively. The full distribution for each scoring function is plotted along the box.

per complex, with the binding affinity and binding probability being predicted by the B2-C classifier and the B2-A regressor, respectively. In all cases, ligands have been positioned in the right pocket. This success is not really surprising because all targets have already been described in the Protein Data Bank in complex with ligands sharing the same pocket and functional effect as the VS hits considered in the current study.

Intriguingly, affinity predictions by both models are remarkably accurate, with median ROC AUC values of 0.763 and 0.740 for binding probability and binding affinity, respectively (Figure 1, Table S1). Boltz-2 failed on only two target sets (CNR1, MTR1A) with ROC AUC values below 0.65 (Table S1). Out of the ten tested rescoring methods, the two Boltz-2 models provide the only two metrics giving distributions being statistically different from those given by the original docking score (*t* test at the 0.05 level; Table S1).

Despite excellent classification accuracy, correlations between predicted and experimental binding affinities of true positives remain very poor, except for the ROCK1 target set

(Figure S1). It remains, however, very intriguing that cofolding succeeds on a data set on which a semiempirical quantum mechanics method (PM6/COSMO2) previously failed.²⁰ We could not observe any relationship between Boltz-2 accuracy of both models (binding affinity, binding probability) and the number of available ChEMBL ligands (binders, nonbinders) and PDB structures, for each of the ten ULVSH targets (Table S2). We therefore checked whether Boltz-2 ranking predictions could have been potentially biased by some target or compound features that the models may have memorized during the training.

Boltz-2 Predictions Are Not Biased by Obvious Compound Similarity or Refolding Failures

One possible explanation for the apparent excellent performance of Boltz-2 in discriminating true from false positives may lie in the high chemical similarity between ULVSH and ChEMBL compounds that Boltz-2 has been trained on. We therefore computed the 2D similarity between all ULVSH compounds and ChEMBL ligands annotated for the same target, regardless of their affinity label (potency, active vs inactive) and kept only the highest similarity value for each ULVSH compound (Figure 2A). Only four of the 943 ULVSH compounds are present in ChEMBL, all from the DRD4 set. For all but the DRD3 set, similarity values are centered on very low numbers (Tanimoto similarity from Morgan-2 fingerprints around 0.30), indicating low chemical similarity using this 2D descriptor between ChEMBL and ULVSH compounds. The highest pairwise similarities are indeed shifted to higher values for the DRD3 set, but we do not suspect any apparent bias that may have spoiled Boltz-2 predictions. The DRD3 set is not the one for which the affinity predictions are the most accurate (Table S1). Moreover, similarity distributions are almost indistinguishable between target sets on which Boltz-2 succeeded (e.g., ADRA2B, CNR2, and ROCK1) and those on which it failed (CNR1, MTR1A; Figure 2A). Lastly, we could not observe an increased accuracy of B2-C and B2-A models applied to subsets of merged ULVSH ligands exhibiting increasingly similar analogues in ChEMBL (Figure S2).

The same 2D similarity analysis was extended to 2.4 million ChEMBL compounds, regardless of their targets, but did not disclose additional information compared with the above analysis (Figure S3). 2D similarities are indeed shifted to higher values (0.5–0.6), but no obvious biases can be noticed. To account for putative 3D similarity that is not detectable from simple 2D fingerprints, we overlaid with ROCS³⁷ the docking pose of each ULVSH hit with the above ChEMBL compounds for which 250 diverse conformers have previously been generated with Omega2.³⁷ Considering the 3D similarity as the sum of shape and pharmacophore similarity (expressed by the TanimotoCombo coefficient), we observe a drastic shift to high 3D similarity (TanimotoCombo > 1.2) between ULVSH and ChEMBL ligands for all but the SGMR2 set (Figure 2B). Given that the SGMR2 set is one for which binary classification of actives vs inactives is correct, a high 3D similarity of ULVSH to ChEMBL compounds does not seem to be mandatory for good affinity predictions.

Moreover, in cases of good binary classification, the 3D similarity distributions are almost identical when considering only true positive or false positive hits (Figure S4). This observation means that both true and false positives have near neighbors in the ChEMBL 3D space. Strikingly, the ULVSH

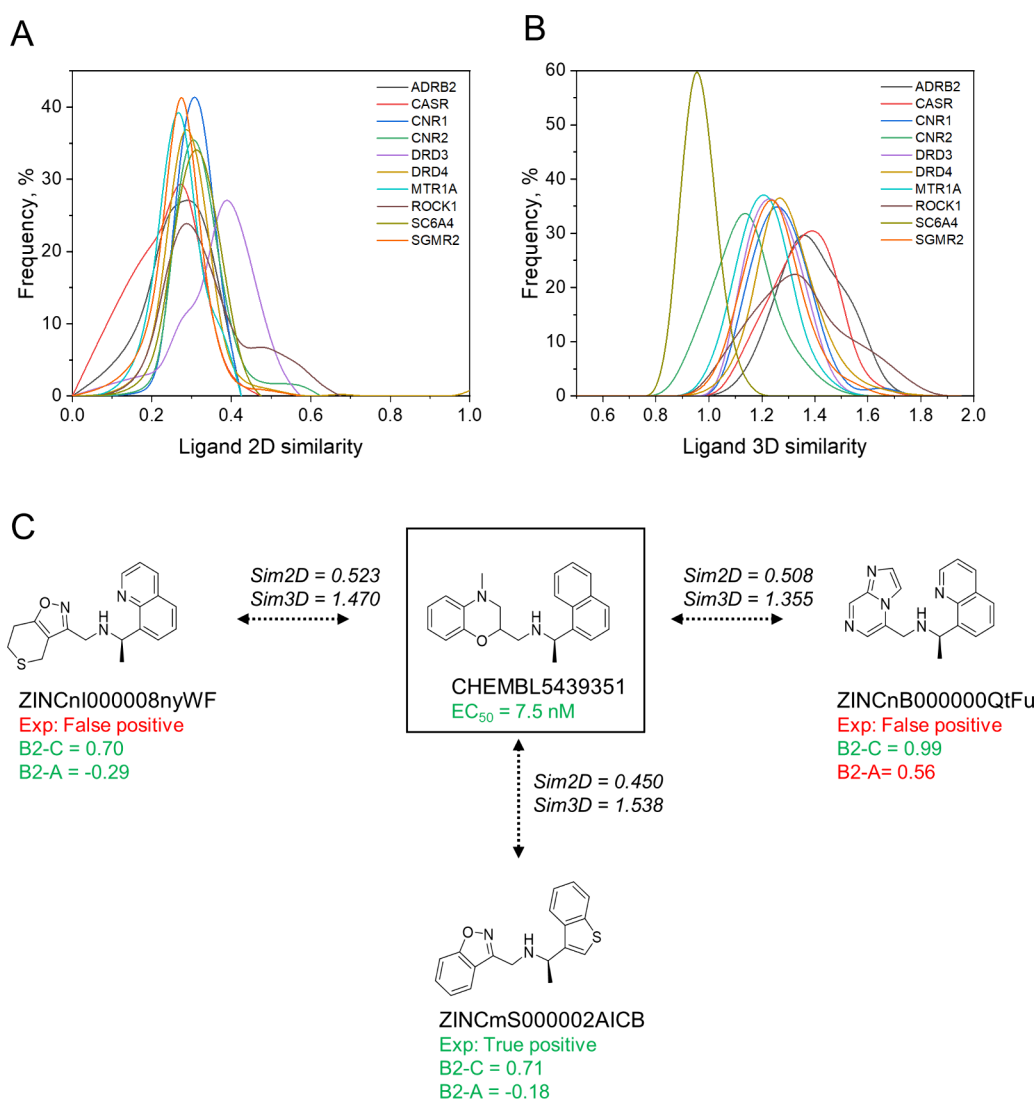


Figure 2. Search for potential biases in Boltz-2 affinity predictions. (A) Highest pairwise 2D similarity (Tanimoto coefficient from Morgan-2 fingerprints) between ULVSH compounds and ChEMBL ligands sharing the same target; (B) highest pairwise 3D similarity (TanimotoCombo coefficient from ROCS overlap) between ULVSH compounds and ChEMBL ligands sharing the same target; (C) Example of three CASR virtual screening hits (ZINC compounds) sharing the same ChEMBL ligand (CHEMBL5439351) as most 3D similar compound, according to OpenEye's ROCS overlay. 2D and 3D pairwise similarities are labeled "2Dsim" and "3DSim", respectively. Experimental affinity labels (true positive, false positive) are taken from *in vitro* functional assays.²⁴ Affinity predictions by Boltz-2 are expressed either as binding probability (B2-C) or affinity values (B2-A) given in $\log_{10}[\text{IC}_{50}, \mu\text{M}]$ unit.

compounds are less similar to each other than to ChEMBL ligands (self 3D similarity of ULVSH hits is centered on values around 0.6–0.7).²⁰

Looking at the available binding data in ChEMBL, we could not find any relationship between the affinity value (label) of those neighbors and that of the most 3D similar ULVSH hits (Figure 2C). Taking the case of CASR hits as an illustrative example, the same ChEMBL potent ligand (CHEMBL5439351) was indeed found as the closest 3D neighbor to two false positives and a true positive ULVSH compound (Figure 2C). Boltz-2 affinity predictions were not influenced by this 3D neighborhood since a false positive was erroneously predicted as CASR binder, whereas the second false positive was correctly predicted as nonbinder (Figure 2C). There is therefore no general trend that we could detect between Boltz-2 performance and possible 3D similarity between training and test compounds.

Since the Boltz-2 refolding (B2-S) and affinity (B2-A, B2-C) prediction models are interdependent,¹¹ we next checked whether the accuracy at which each target has been refolded may influence affinity predictions. Again, no obvious relationships could be found, binding site-lining amino acids being very well modeled in eight out of ten cases (Figure 3). Higher deviations are, however, observed for CASR and SC6A4 predicted structures but did not preclude a good binary classification of corresponding ULVSH compounds. These discrepancies are likely due to the heterogeneity of available PDB templates, for which several activation states are available. Hence, SC6A4 predicted structures are in the outward open conformation whereas the structure used for docking is in the inward open conformation,⁴² thereby explaining the 3 Å RMSD shift in active site residues (Figure 3). Since SC6A4 inhibitors may bind to both states,⁴² the consequences may be limited in our study. More questionable are some predicted CASR structures which were much closer to the symmetric

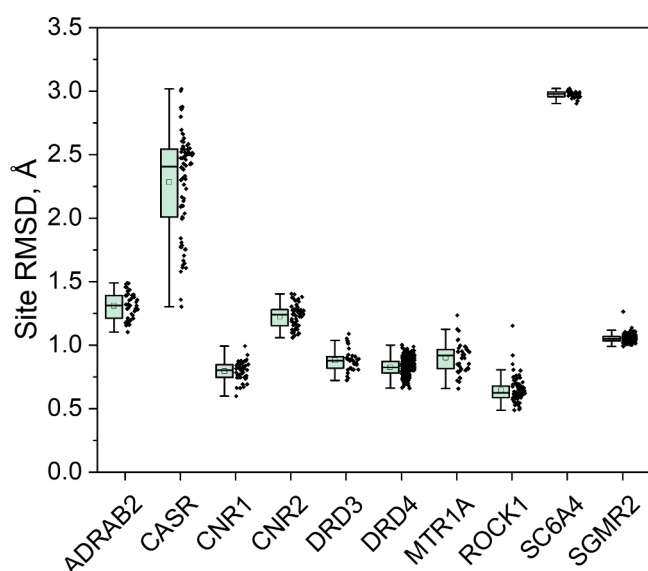


Figure 3. Root-mean square deviations (binding site heavy atoms) of Boltz-2 predicted structure to the experimental structure used for docking. The boxes delimit the first and the third quartiles, and the whiskers delimit the 1.5 standard deviation. The median and the mean scores are indicated by a horizontal line and a square located in each box, respectively. The full distribution for each RMSD calculation is plotted along the box.

inactive state⁴³ than to the asymmetric active state⁴⁴ to which the CASR positive allosteric modulators (PAMs) have been originally docked.³⁰ This discrepancy does not prevent Boltz-2 from accurately distinguishing true from false positives refolded in a wrong receptor activation state.

Boltz-2 Affinity Predictions Are Tolerant to a Poor Pose Quality

We next looked at the accuracy of the B2-S model to reproduce the docking pose responsible for the selection of ULVSH compounds for experimental validation. Although we cannot exclude that some original docking poses are incorrect (notably for false positives), we are confident about the quality of true positive poses given the postprocessing constraints forcing hits to verify key interactions shared by true actives, the confirmed binding mode of a few hits by structure determination of the corresponding complexes, and successful structure-guided hit optimization reports.^{21–30} Surprisingly, the Boltz-2 poses, despite being located in the right pocket and verifying, for most true positives, one or two expected key interactions (Figure S5), deviate significantly from the original docking poses considered here as the ground truth (Figure 4). In most cases, poses from false positives tend to deviate more from docking poses than those from true positives (e.g., DRD4, MTR1A, ROCK1). However, no stringent differences are observed for most targets (e.g., ADRA2B, CASR, CNR1, CNR2, DRD3, SC6A4, and SGMR2). Strikingly, the affinity

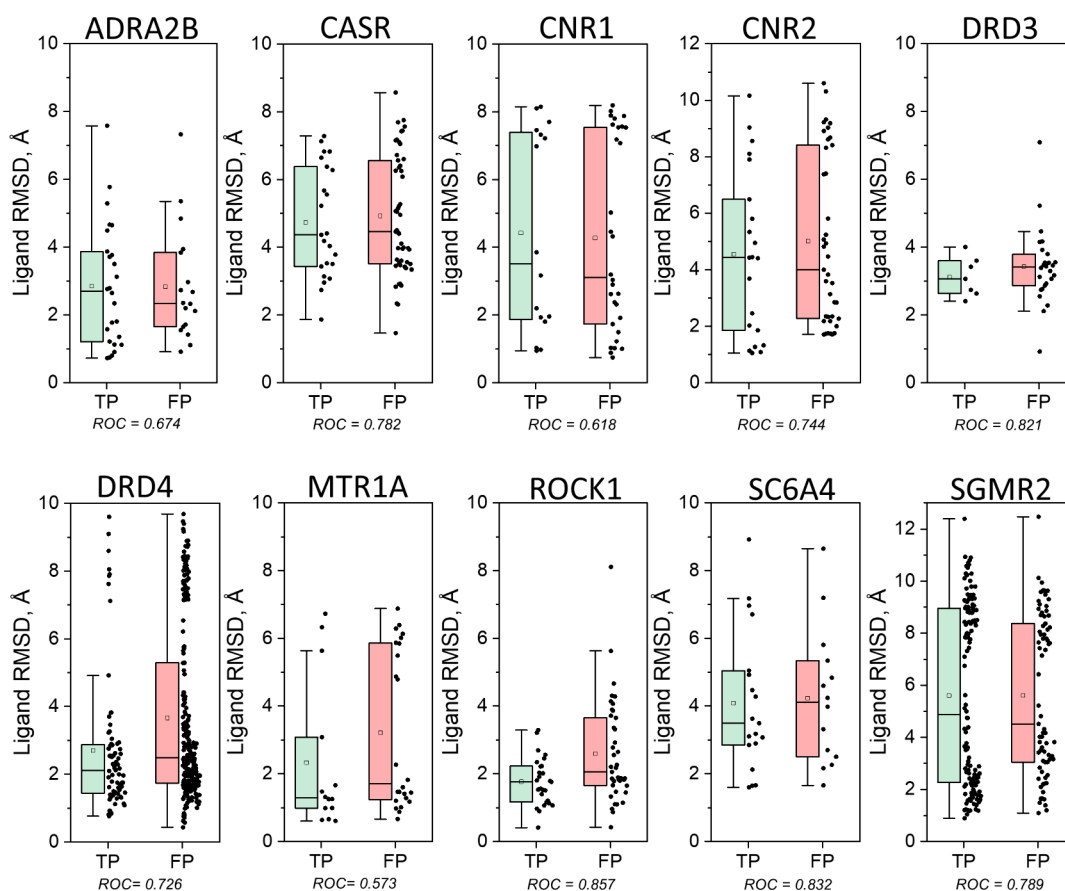


Figure 4. Root-mean square deviations (heavy atoms) of Boltz-2 predicted ULVSH ligand structures (TP, true positives; FP, false positives) to the original docking poses. The boxes delimit the first and the third quartiles, and the whiskers delimit the 1.5 standard deviation. The median and the mean scores are indicated by a horizontal line and a square located in each box, respectively. The full distribution for each RMSD calculation is plotted along the box. Areas under the ROC curves are indicated for each target by using the Boltz-2 binding probability as a metric.

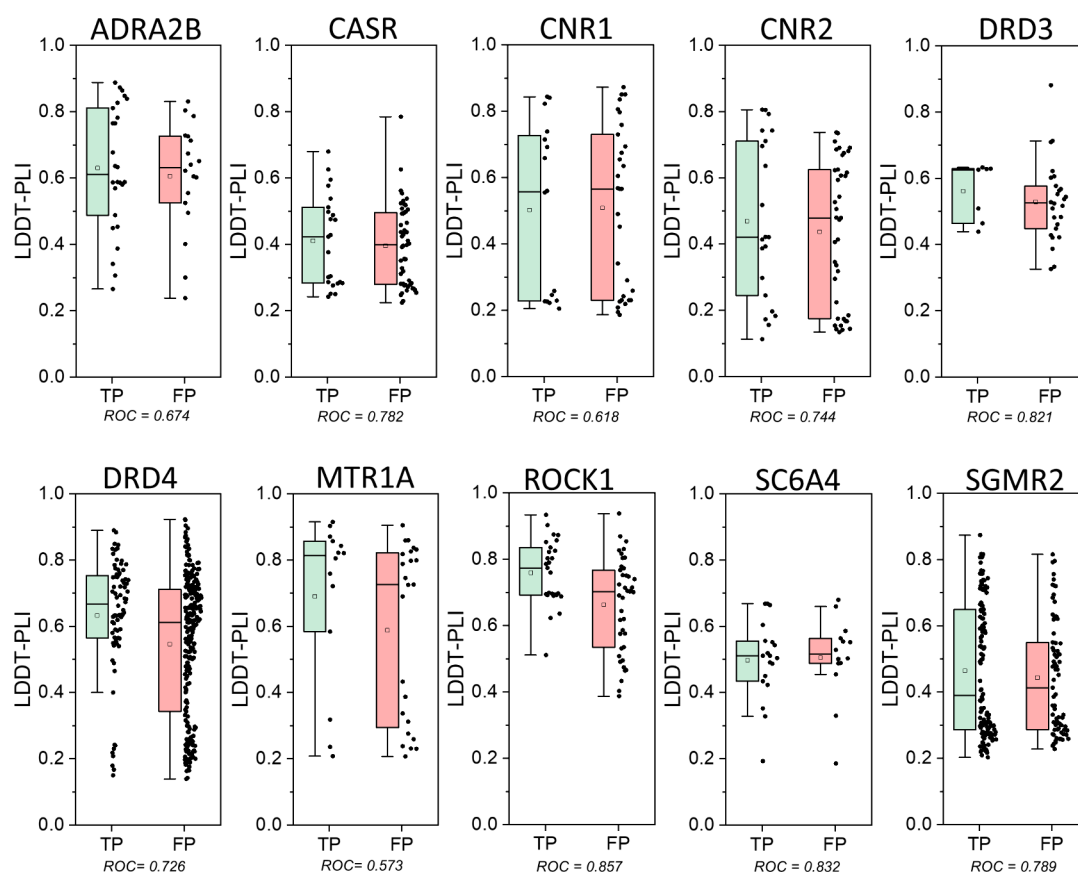


Figure 5. Local distance difference test at the protein–ligand interface considering Boltz-2 complexes and original docking protein–ligand complexes (TP, true positives; FP, false positives). The boxes delimit the first and the third quartiles, and the whiskers delimit the 1.5 standard deviation. The median and the mean scores are indicated by a horizontal line and a square located in each box, respectively. Areas under the ROC curves are indicated for each target using the Boltz-2 binding probability as a metric.

predictions are not correlated with pose accuracy. Poor pose accuracy (e.g., CASR, SC6A4, SGMR2) does not prevent Boltz-2 from efficiently discriminating active from inactive compounds (Figure 4).

Since the RMSD is a global metric that does not explicitly consider conservation of protein–ligand interactions, we next applied the local distance difference test of protein–ligand interactions (LDDT-PLI)⁴⁵ between Boltz-2 and protein–ligand complexes obtained by docking. In this test that does not require a prior alignment of protein–ligand structures to be compared, the LDDT-PLI metric is defined as the average of four fractions of pairwise distances that exhibit a difference less than 0.5, 1.0, 2.0, and 4.0 Å respectively. Following a recent report,⁴¹ we assumed that protein–ligand interactions are conserved when the LDDT-PLI value is higher than 0.80. Again, we could not find any relationship between the LDDT-PLI distributions and binding probability predictions. Despite conservation of protein–ligand interactions between Boltz-2 and docking poses is rarely observed, the B2-C classifier manages to efficiently discriminate true from false positive hits in most cases (Figure 5).

An illustrative example is given by the SC6A4 set, for which only 50% of intermolecular interactions are conserved between docking and Boltz-2 poses (Figure 5). The ability of the Boltz-2 affinity prediction models to perform very well on low-resolution poses, for which many expected protein–ligand interactions are missing, raises concerns about the real features governing these predictions. To assess whether Boltz-2 is just

memorizing compound and/or target features it has been trained on, we designed a series of biologically meaningful tests to challenge its affinity predictions in a straightforward manner.

Adversarial Challenges Question the Generalizability of Boltz-2 Affinity Predictions

A recent report highlighted the lack of sensitivity of AF3 protein–ligand cofolding to biologically meaningful challenges (amino acid mutations) drastically affecting the nature of a binding pocket.¹⁷ Hence, AF3 binding poses remained largely unaffected by drastic binding site mutations, contradicting elementary physical principles and suggesting that AF3 has not yet fully learned the major laws of protein–ligand recognition. Following the same idea, we generated site-directed mutants for each of the ten ULVSH targets (Table S3), known to prevent ligand binding by suppressing the most important polar interactions^{21,23,28,44,46–50} shared by almost all true ligands (Figure S6). For example, the key aspartic acid at position 3.32⁵¹ of monoamine G protein-coupled receptors (residue D92 in ADRA2B, D110 in DRD3, D115 in DRD4) was mutated to alanine, a mutation known to suppress ligand binding to these receptors.⁵² Likewise, the very important hinge residue in the ROCK1 kinase (Met153) was mutated to proline to remove the key anchoring hydrogen-bond to the hinge backbone nitrogen atom (Figure S6). Note that a perfect model would be expected to classify all these adversarial inputs as nonbinders. Such fixed predictions should lead to an ROC AUC value close to 0.5 on the original labels, regardless of the class balance.

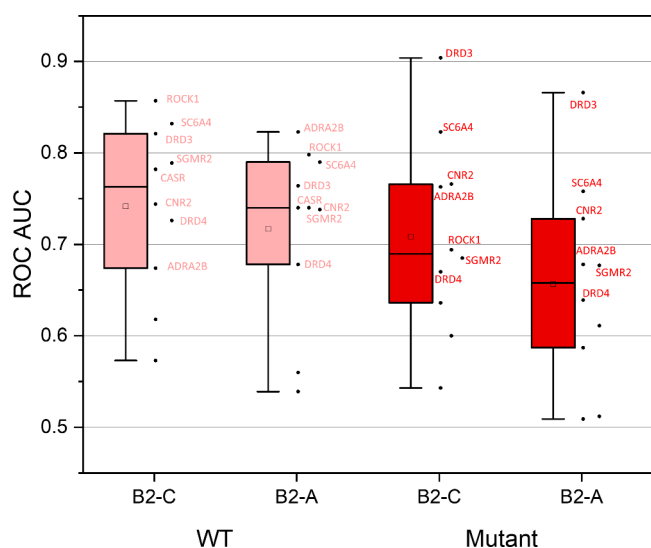


Figure 6. Area under the ROC curves (ROC AUC) for classifying true active and true inactive compounds for ten target sets upon cofolding with wild type (WT) and mutated (mutant) targets. Predictions from Boltz-2 are labeled “B2-C” for binding probability and “B2-A” for binding affinity. The boxes delimit the first and the third quartiles, and the whiskers delimit the 1.5 standard deviation. The median and the mean scores are indicated by a horizontal line and a square located in each box, respectively. The full distribution for each scoring function is plotted along the box. Targets with acceptable affinity predictions (ROC AUC > 0.65) are explicitly labeled.

ULVSH compounds were thus cofolded again with the site-directed mutants, and activity labels were predicted. Despite yielding lower averaged ROC AUC values than those obtained with wild type proteins, Boltz-2 still manages to efficiently discriminate the activity labels in most cases with predictions not statistically different from those achieved with the wild type targets (Figure 6, Table S4).

Interestingly, Boltz-2 poses to site-directed mutants deviate significantly from those observed with the wild type receptor, the deviations being equally observed for active and inactive compounds (see an example of the ADRA2B ligand in Figure S7). If the poses to wild type and mutants had been very similar, we could have understood that the relative ranking remains unchanged for a target, with the same interactions to the mutated amino acids being systematically ignored for every compound. However, since this is not the case, obtaining comparable prediction accuracy on wild type and mutated targets from different binding poses remains hard to understand from the physics of intermolecular interactions.

To check whether Boltz-2 binding affinity models are really focusing on the binding pocket and not predicting affinities from long-range protein–ligand patterns, as often observed for machine learning scoring functions,⁵³ we performed a series of deeper analyses. More precisely, we generated a series of mutations (wild-type to alanine) for each protein, specifically targeting either the binding site or the outer shell of each target, with an increasing mutation rate (0.02, 0.03, and 0.1; Table S5). Depending on the full sequence length and the size

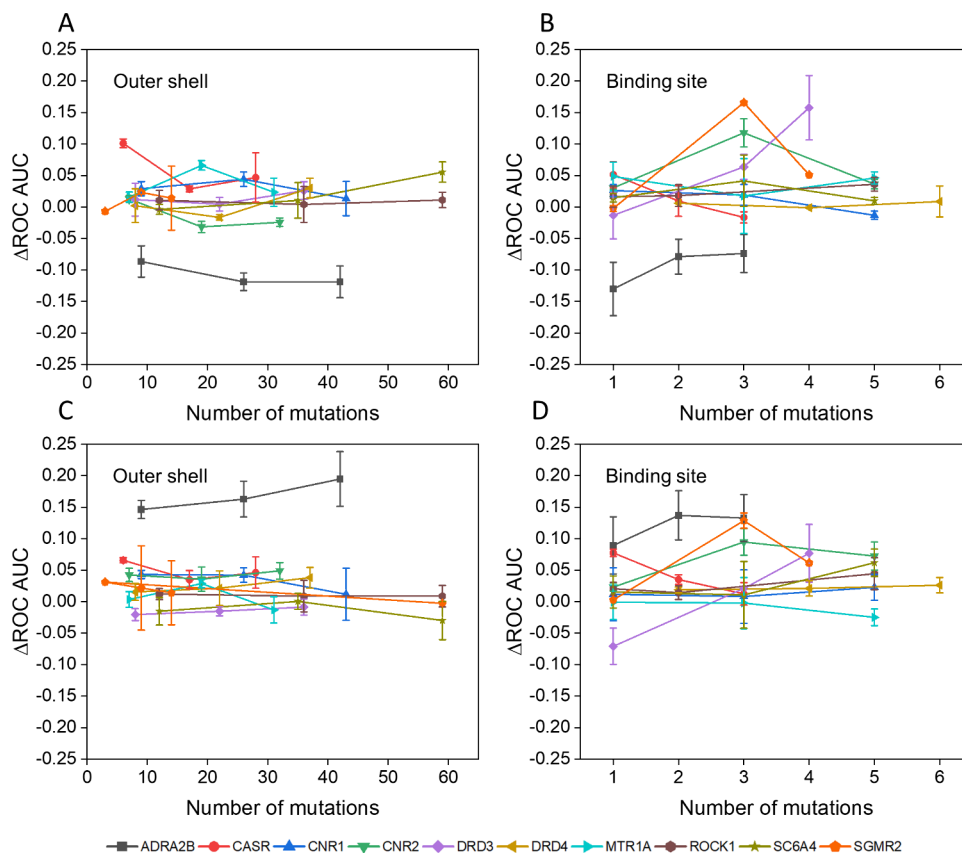


Figure 7. Variation of the area under the Δ ROC curves (Δ ROC AUC = $\text{ROC AUC}_{\text{wt}} - \text{ROC AUC}_{\text{mutant}}$) for classifying true active and true inactive compounds for ten target sets upon cofolding with target sequences submitted to an increasing number of mutations at randomly chosen positions. Mutations are directed to either the outer shell (A, C) or the binding site (B, D). Metrics used for binary classification are binding probability (A, B) and binding affinity (C, D). ROC AUC and standard deviations were determined from computations performed in triplicate.

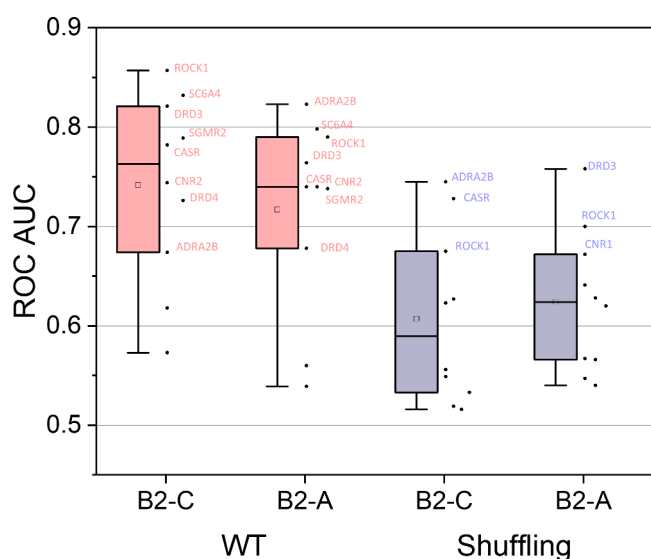


Figure 8. Area under the ROC curves (ROC AUC) for classifying true active and true inactive compounds for ten target sets upon cofolding with a shuffled target. Predictions from Boltz-2 are labeled “B2-C” for binding probability and “B2-A” for binding affinity. The boxes delimit the first and the third quartiles, and the whiskers delimit the 1.5 standard deviation. The median and the mean scores are indicated by a horizontal line and a square located in each box, respectively. The full distribution for each scoring function is plotted along the box. Targets with acceptable affinity predictions (ROC AUC > 0.65) are labeled.

of the binding site, the number of mutations ranges from 1 to 6 for the binding site and from 3 to 59 for the outer shell (Table S5).

As is to be expected, increasing the number of mutations outside the binding site did not affect the binary classification of true and false positives with respect to that obtained with the wild-type sequences, whatever the metric used by Boltz-2 (binding probability or binding affinity; Figure 7A,C). An even slightly increased classification accuracy was observed for the ADRA2B set when the binding probability was used as a descriptor (Figure 7A). Conversely, the classification was worse for the same set using the binding affinity values predicted by the B2-A model (Figure 7C). For the nine remaining targets, variations were limited and within ± 0.05 units of the original ROC AUC value obtained for the wild-type sequences.

Surprisingly, despite variations of larger intensities, the effect of mutating the ligand-binding site remained quite limited and not proportional to the number of mutations in most cases (Figure 7B,D). The only clear effect was observed for the DRD3 set, for which, as would be expected, the ROC AUC value decreases linearly with the number of mutations when binding probability was retained as a descriptor (Figure 7B). For other sets, variations of the ROC AUC are limited to ± 0.10 units, which means that the classification accuracy of the two affinity prediction models using mutated binding site sequences still remains better than that obtained by all originally tested scoring functions on the wild-type structure (recall Figure 1). Expressing the results as a function of the mutation rate leads to the same conclusion (Figure S8). The current results are in line with the data presented above on user-selected key binding site mutations and suggest that Boltz-

2 affinity predictions remain relatively insensitive to biologically significant binding site mutations.

In order to present even more significant challenges, we designed another test in which targets were shuffled, assigning ULVSH compounds to a totally unrelated target (Table S6). ULVSH compounds were refolded with the wrong target sequences, still keeping the affinity and activity labels associated with the original target. Since each target exchange is assumed to prohibit the binding of all ULVSH ligands, labels should be “nonbinder” for all compounds and the expected binary classification should be close to that reached with a random classification (ROC AUC = 0.50) for every target.

Reassuringly, average ROC AUC values decrease significantly to almost random values (ROC AUC < 0.6) but only for five out of ten targets when the B2-C classifier was used, and for four of them when the B2-A regressor was applied to output continuous values (Figure 8). For five targets (ADRA2B, CASR, ROCK1, CNR1, and DRD3), binding probability and affinity predictions of the corresponding ligands were, however, almost target-independent (Figure 8).

Although we acknowledge that Boltz-2 has not been trained on extreme and biologically irrelevant situations, cofolding DRD3 ligands with a single alanine amino acid as the target sequence still led to good binary classification of binders vs nonbinders (Table S7), evidencing that Boltz-2 affinity predictions are in this case exclusively ligand-dependent.

CONCLUSIONS

This report challenges the recently described Boltz-2 cofolding and binding affinity models to efficiently distinguish between true and false positive compounds issued from ten ultralarge-virtual-screening hit lists. Our study highlights contradictory conclusions with respect to the relevance of the obtained results. On the positive side, Boltz-2, used out of the box, remarkably classifies experimentally active and inactive compounds for eight out of ten ULVSH targets, despite the low 2D chemical similarity of tested ligands to known compounds on which Boltz-2 has been trained. The performance is remarkable, given the difficulty of the ULVSH data set on which even rigorous physics-based scoring functions previously failed.²⁰ Moreover, a target shuffling exercise, in which a wrong target was cofolded with the right ligands, results in a random classification performance for a majority of targets. Last, systematic mutation of the outer binding site shell also leads to the expected unchanged classification accuracy, indicating that Boltz-2 is really focusing at the protein–ligand interface.

On the negative side, the affinity predictions are insensitive to the quality of ligand poses, which often deviate significantly from the expected ground-truth. Moreover, predictions remain globally unchanged after mutating binding site residues and even after exchanging targets in three cases. Boltz-2 predictions are known to be tightly dependent on the availability of closely related PDB templates and known ligands. In the ULVSH test set, the 3D structures of all targets (holo- and apoforms) are already part of the training data. Moreover, we showed that high 3D ligand similarity (shape and pharmacophoric patterns) to known ChEMBL ligands is observed for almost all ULVSH compounds whatever their label (binder, nonbinder). These data clearly put Boltz-2 in the best possible conditions for predicting binding affinities for ULVSH compounds. However, we could not find any obvious bias when comparing predicted and experimental binding affinities, solely based on ligand 3D

similarity to known ligands of the same target. We could not rationally explain how Boltz-2 manages to achieve good binary classification of binders vs nonbinders, in the absence of statistical correlation between predicted and experimental binding affinities, and from poor ligand poses. Mutation challenges specifically directed at the binding pocket of targets already present in the training data clearly suggest a dominant role of ligand information (3D shape and pharmacophoric patterns) in Boltz-2 predictions. However, we could not evidence any significant 2D/3D ligand data leakage that may have influenced affinity predictions.

In the absence of true prospective validations of Boltz-2 usage to select hit compounds from docking screens, we therefore advise remaining very skeptical with respect to affinity predictions outputted by this AI cofolding program relying on a huge number of hyperparameters. Our study emphasizes the need to carefully define the applicability domain of AI models, to make explicitly precise their limitations, and whenever possible to prospectively apply the model with limited risks of data leakage.

■ ASSOCIATED CONTENT

Data Availability Statement

Atomic coordinates (protein, binding site, ground-truth ligand, docking poses; MOL2 file format), in vitro data (tsv format) from the ULVSH set are freely available at <https://drugdesign.unistra.fr/lab/downloads/ULVSH.tgz>. All Boltz-2 input (.yaml files) and output data (.pdb, .json and .npz files) are available as a compressed archive (tgz format) at zenodo with the DOI 10.5281/zenodo.16946890. Boltz-2 is available under MIT License at <https://github.com/jwohlwend/boltz>. All scripts to compare Boltz-2 to docking poses (ligands, protein, protein–ligand complexes) are available at https://github.com/LIT-CCM-lab/Boltz-2_performance.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.5c02630>.

Area under the ROC curve for classifying ULVSH compounds as active or inactive by ten different scoring functions; ChEMBL and PDB status of ULVSH targets; mutants evaluated for cofolding and binding affinity predictions; area under the ROC curve for classifying compounds as active or inactive after site-directed mutagenesis or target shuffling; number of mutations considered in the binding site and the outer shell of the ten ULVSH targets; target shuffling for cofolding and affinity predictions; area under the ROC curve for discriminating 33 DRD3 virtual screening hits (seven true positives, 26 false positives) upon cofolding with various target sequences; Boltz-2 predicted vs experimental binding affinities, expressed as $\log_{10}[\text{IC}_{50} \mu\text{M}]$, for seven out of ten ULVSH target sets; area under the ROC curve for discriminating active from inactive ULVSH ligand subsets (all targets merged), selected according to the most 2D similar ChEMBL ligands sharing the same target; highest pairwise 2D similarity (Tanimoto coefficient from Morgan-2 fingerprints) between ULVSH compounds and 2.4 million ChEMBL ligands of known target; 3D similarity, expressed by the ROCS TanimotoCombo coefficient, between ULVSH and ChEMBL ligands sharing the same target; percentage of Boltz-2 poses verifying key protein–ligand

interactions; adversarial challenge designed to assess Boltz-2 affinity prediction; root-mean-square deviation of ADRA2B ligand heavy atoms cofolded with site-directed mutants with respect to those cofolded with the wild-type receptor; variation of the area under the ROC curves ($\text{DROC AUC} = \text{ROC AUC}_{\text{wt}} - \text{ROC AUC}_{\text{mutant}}$) for classifying true active and true inactive compounds for ten target sets upon cofolding with target sequences submitted to an increasing mutation rate (0.02, 0.06 and 0.10) at randomly chosen positions (PDF)

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Author Contributions

G. Bret: installation of software packages and libraries, computation of ligand pairwise similarities; F. Sindt: Boltz-2 calculations; D. Rognan: conceptualization of the study, methodology, Boltz-2 calculations, data analysis, writing the manuscript.

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The authors declare the following competing financial interest(s): D.R. is cofounder and shareholder of BIODOL Therapeutics. The other authors declare no competing interests.

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